

§7.8 - Improper Integrals

Remark

For

$$\int_a^b f(x) \, dx$$

to exist, we require:

1. $[a, b]$ to have finite length
2. f is defined everywhere on $[a, b]$, but f need not be continuous on $[a, b]$

Remark (From §2.6)

If $n > 0$

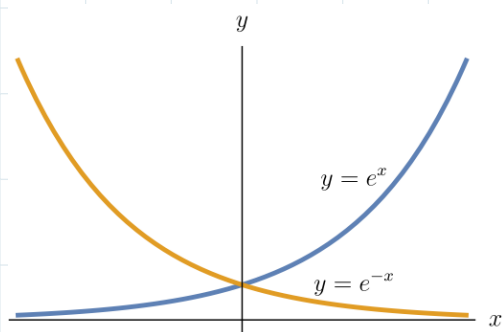
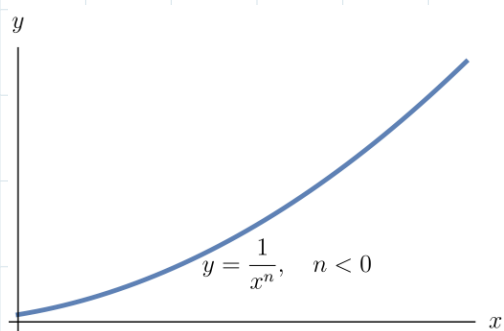
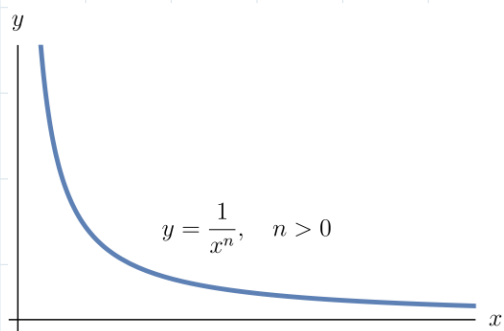
$$\lim_{x \rightarrow \infty} \frac{1}{x^n} = 0$$

and $n < 0$ implies

$$\lim_{x \rightarrow \infty} \frac{1}{x^n} = \infty.$$

While

$$\lim_{x \rightarrow \infty} e^x = \infty \quad \text{and} \quad \lim_{x \rightarrow \infty} e^{-x} = \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0.$$



Definition #1 (Improper Integral I)

If

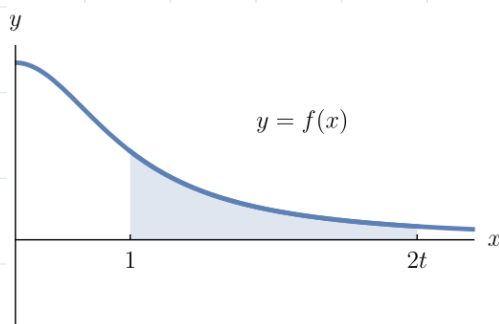
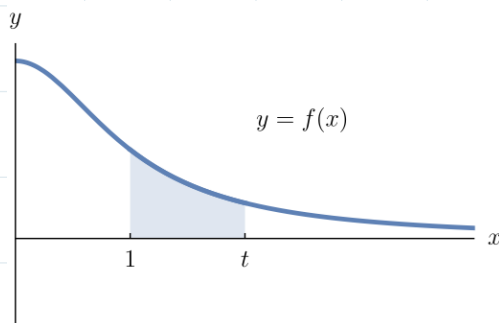
$$\int_a^t f(x) \, dx$$

exists for all $t \geq a$ then

$$\int_a^\infty f(x) \, dx = \lim_{t \rightarrow \infty} \int_a^t f(x) \, dx$$

is the **improper integral** of f over $[a, \infty)$.

Proper integrals require a bounded interval.



Cheat by using the F.T.C. to extend the interval's length,

$$\lim_{t \rightarrow \infty} \int_1^t f(x) dx = \lim_{t \rightarrow \infty} (F(t) - F(1)).$$

Definition #2 (Improper Integral I)

If

$$\int_t^b f(x) \, dx$$

exists for all $t \leq b$ then

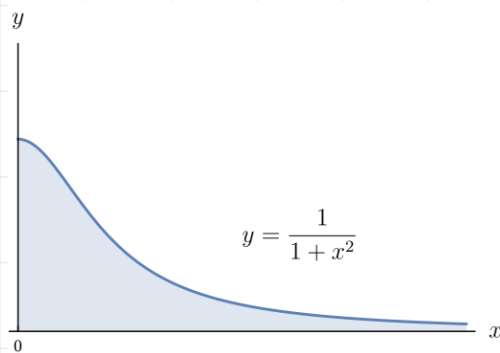
$$\int_{-\infty}^b f(x) \, dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) \, dx$$

is the **improper integral** of f over $(-\infty, b]$.

Example #1

Find

$$\int_0^{\infty} \frac{1}{x^2 + 1} dx.$$

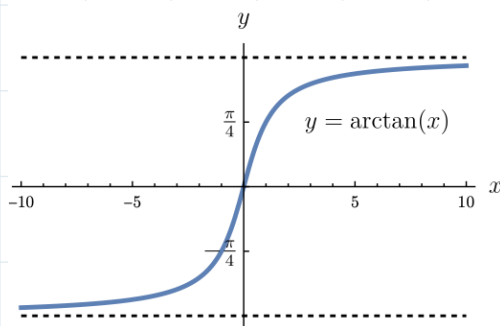
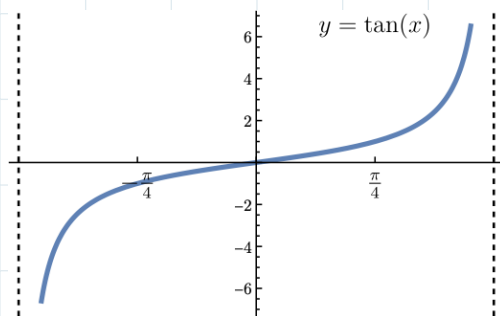


Here

$$\int \frac{1}{1+x^2} dx = \arctan(x) + C$$

so

$$\lim_{t \rightarrow \infty} \int_0^t f(x) dx = \lim_{t \rightarrow \infty} (\arctan(t) - \arctan(0)).$$



With

$$\lim_{t \rightarrow \infty} \arctan(t) = \frac{\pi}{2} \quad \text{and} \quad \arctan(0) = 0$$

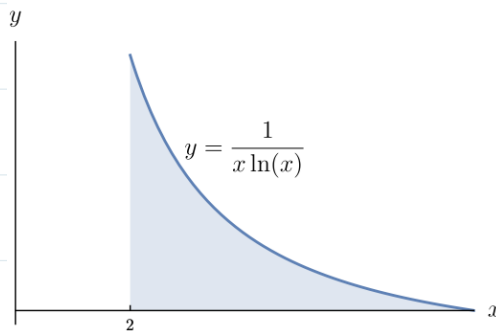
we get

$$\int_0^{\infty} \frac{1}{1+x^2} dx = \frac{\pi}{2}.$$

Example #2

Find

$$\int_2^{\infty} \frac{1}{x \ln(x)} dx.$$



Let

$$u = \ln(x) \quad \text{and} \quad du = \frac{1}{x} dx$$

then

$$\int \frac{1}{x \ln(x)} dx = \int \frac{1}{u} du = \ln(u) + C = \ln(\ln(x)) + C$$

so

$$\begin{aligned} \int_2^{\infty} \frac{1}{x \ln(x)} dx &= \lim_{t \rightarrow \infty} \int_2^t \frac{1}{x \ln(x)} dx \\ &= \lim_{t \rightarrow \infty} \ln(\ln(x)) \Big|_{x=2}^t \\ &= \lim_{t \rightarrow \infty} (\ln(\ln(t)) - \ln(\ln(2))) \\ &= \infty \end{aligned}$$

since

$$\lim_{t \rightarrow \infty} \ln(t) = \infty.$$

Definition #3

Suppose

$$I = \int_a^{\infty} f(x) \, dx \in \mathbf{R}.$$

Then the improper integral **converges**.

If $I \notin \mathbf{R}$, then the improper integral **diverges**.

Warning

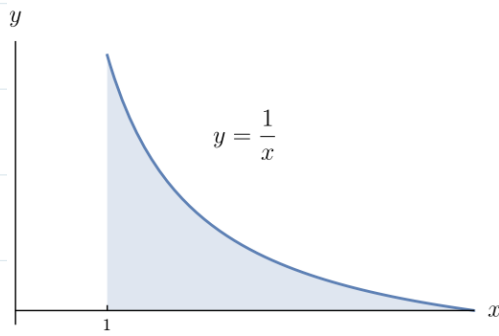
We can have

$$\lim_{x \rightarrow \infty} f(x) = 0 \quad \text{and} \quad \int_0^{\infty} f(x) \, dx = \infty.$$

Example #3

Find

$$\int_1^{\infty} \frac{1}{x} dx.$$



Here

$$\begin{aligned}\int_1^{\infty} \frac{1}{x} dx &= \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx \\ &= \lim_{t \rightarrow \infty} \ln(x) \Big|_{x=1}^t \\ &= \lim_{t \rightarrow \infty} (\ln(t) - \ln(1)) \\ &= \infty\end{aligned}$$

since

$$\lim_{t \rightarrow \infty} \ln(t) = \infty.$$

Thus

$$\int_1^{\infty} \frac{1}{x} dx = \infty$$

but

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0.$$

Definition #4 (Improper Integral I)

If

$$I_1 = \int_{-\infty}^a f(x) \, dx \quad \text{and} \quad I_2 = \int_a^{\infty} f(x) \, dx$$

converge then

$$\int_{\mathbf{R}} f(x) \, dx = \int_{-\infty}^{\infty} f(x) \, dx = I_1 + I_2.$$

Warning

The terms

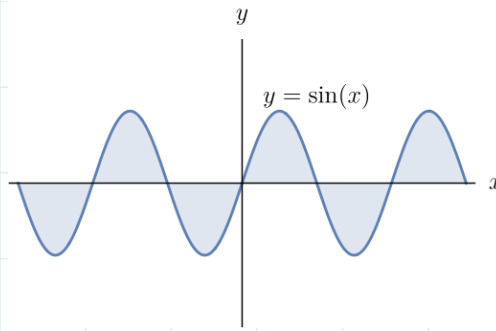
$$\int_{\mathbf{R}} f(x) \, dx \quad \text{and} \quad \lim_{t \rightarrow \infty} \int_{-t}^t f(x) \, dx$$

need **not** be equal.

Example #4

Find

$$\int_{\mathbf{R}} \sin(x) \, dx.$$



Notice

$$\lim_{t \rightarrow \infty} \int_0^t \sin(x) dx = \lim_{t \rightarrow \infty} (-\cos(x)) \Big|_{x=0}^t \quad *$$

diverges since

$$\lim_{t \rightarrow \infty} \cos(t) \text{ d.n.e.}$$

However sine is odd,

$$\int_{-t}^t \sin(x) dx = 0 \quad \text{for all } t > 0$$

$$\Rightarrow \lim_{t \rightarrow \infty} \int_{-t}^t \sin(x) dx = 0$$

but

$$\int_0^{\infty} \sin(x) dx \text{ diverges}$$

$$\Rightarrow \int_{\mathbb{R}} \sin(x) dx \text{ diverges}$$

by *.

Note:

$\int_{\mathbb{R}} f(x) dx$ converges $\Rightarrow \int_{\mathbb{R}} f(x) dx = \lim_{t \rightarrow \infty} \int_{-t}^t f(x) dx$
since $\int_{\mathbb{R}} f(x) dx$ converges for all "paths" to ∞ .

Definition #5 (Improper Integral II)

Let f be a continuous function on $(a, b]$ with

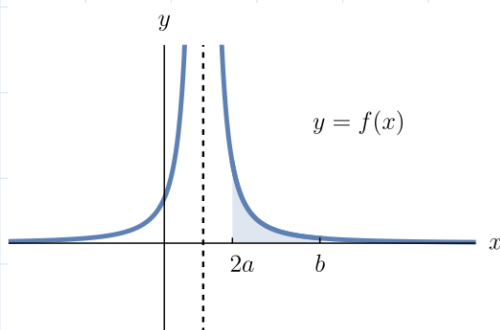
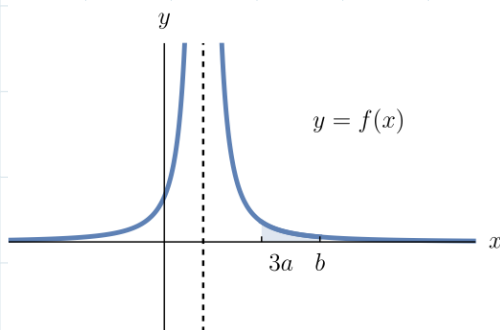
$$\lim_{x \rightarrow a^+} f(x) = \pm\infty.$$

Then

$$\int_a^b f(x) \, dx = \lim_{t \rightarrow a^+} \int_t^b f(x) \, dx$$

is the **improper integral** of f over $[a, b]$.

Proper integrals require a bounded function.



Cheat by using the F.T.C. to integrate near the unbounded growth

$$\lim_{t \rightarrow a^+} \int_t^b f(x) dx = \lim_{t \rightarrow a^+} (F(b) - F(t))$$

Example #5

Find

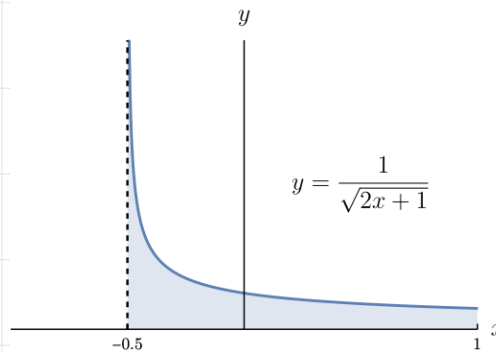
$$\int_{-1/2}^1 \frac{1}{\sqrt{2x+1}} dx.$$

Here

$$f(x) = \frac{1}{\sqrt{2x+1}}$$

is undefined at $x = -1/2$ and

$$\lim_{x \rightarrow -1/2^+} f(x) = \infty.$$



Now

$$\int (2x+1)^{-1/2} dx = (2x+1)^{1/2} + C$$

gives

$$\lim_{t \rightarrow -1/2^+} \int_t^1 f(x) dx = \lim_{t \rightarrow -1/2^+} ((2+1)^{1/2} - (2t+1)^{1/2})$$

and

$$\lim_{t \rightarrow -1/2^+} (2t+1)^{1/2} = 0$$

so

$$\Rightarrow \int_{-1/2}^1 (2x+1)^{-1/2} dx = \sqrt{3}.$$

Definition #6 (Improper Integral II)

Let f be a continuous function on $[a, b)$ with

$$\lim_{x \rightarrow b^-} f(x) = \pm\infty.$$

Then

$$\int_a^b f(x) \, dx = \lim_{t \rightarrow b^-} \int_a^t f(x) \, dx$$

is the **improper integral** of f over $[a, b]$.

Definition #7 (Improper Integral II)

Let f be a continuous function on $[a, c) \cup (c, b]$ with f unbounded at $c \in (a, b)$.

If

$$I_1 = \int_a^c f(x) \, dx \quad \text{and} \quad I_2 = \int_c^b f(x) \, dx$$

converge then the **improper integral**

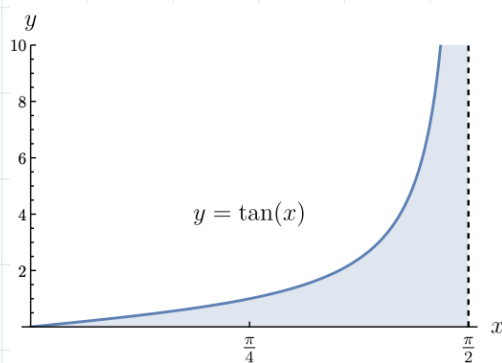
$$\int_a^b f(x) \, dx = I_1 + I_2$$

converges.

Example #6

Find

$$\int_0^{\pi/2} \tan(t) \, dt.$$



With

$$\begin{aligned}
 u &= \cos(x) \quad du = -\sin(x) \, dx \\
 \int \tan(x) \, dx &= \int \frac{\sin(x)}{\cos(x)} \, dx \\
 &= \int -\frac{1}{u} \, du \\
 &= -\ln|\cos(x)| + C \\
 &= \ln|\sec(x)| + C
 \end{aligned}$$

Now

$$\lim_{t \rightarrow \pi/2^-} \int_0^t f(x) \, dx = \lim_{t \rightarrow \pi/2^-} (-\ln|\cos(t)| - (-\ln|\cos(0)|))$$

and

$$\begin{aligned}
 \lim_{t \rightarrow \pi/2^-} \cos(t) &= 0 \quad \lim_{x \rightarrow 0} \ln|x| = -\infty \\
 \Rightarrow \lim_{t \rightarrow \pi/2^-} -\ln|\cos(t)| &= \infty.
 \end{aligned}$$

Hence

$$\int_0^{\pi/2} \tan(t) dt = \infty.$$

Example #7 (Two Types)

Find

$$\int_{\mathbf{R}} \frac{1}{x^2} dx.$$

We have

$$\int x^{-2} dx = -\frac{1}{x} + C$$

$$\lim_{t \rightarrow \infty} \int_1^t f(x) dx = \lim_{t \rightarrow \infty} \left(-\frac{1}{t} + 1 \right) = 1$$

since

$$\lim_{x \rightarrow \infty} -\frac{1}{x} = 0$$

$$\Rightarrow \int_1^{\infty} x^{-2} dx = 1$$

Now

$$\lim_{t \rightarrow 0^+} \int_t^1 f(x) dx = \lim_{t \rightarrow 0^+} \left(-1 + \frac{1}{t} \right)$$

but

$$\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty$$

$$\Rightarrow \int_0^1 \frac{1}{x^2} dx = \infty.$$

Hence

$$\int_0^{\infty} \frac{1}{x^2} dx = \infty \Rightarrow \int_{\mathbb{R}} \frac{1}{x^2} dx \text{ diverges}$$

Theorem #1 (Comparison Test)

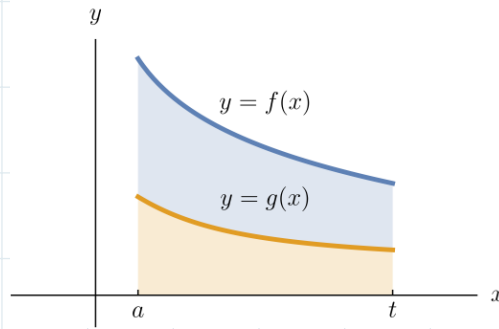
Suppose f and g are continuous on the interval $[a, \infty)$ where

$$0 \leq g(x) \leq f(x)$$

for all $x \geq a$.

Then

1. If $\int_a^\infty f(x) \, dx$ converges then $\int_a^\infty g(x) \, dx$ converges.
2. If $\int_a^\infty g(x) \, dx$ diverges then $\int_a^\infty f(x) \, dx$ diverges.



It is essential that our functions are nonnegative,

$$0 \leq g(x) \leq f(x).$$

If the smaller integral diverges, so must the larger integral.

If the bigger integral converges, so must the smaller integral.

Example #8

Find

$$\int_1^{\infty} \frac{\ln^2(x)}{x} dx$$

Let

$$f(x) = \frac{\ln^2(x)}{x} \quad g(x) = \frac{1}{x}$$

then

$$0 < g(x) \leq f(x)$$

for $x \geq e$ since

$$\ln(x) \geq 1 \quad \text{when } x \geq e$$

With

$$\int_e^{\infty} \frac{1}{x} dx = \infty \Rightarrow \int_e^{\infty} \frac{\ln^2(x)}{x} dx = \infty$$

by the Comparison Test.

Since

$$\int_e^{\infty} \frac{\ln^2(x)}{x} dx = \infty$$

it follows that

$$\int_1^{\infty} \frac{\ln^2(x)}{x} dx = \infty.$$

Or show the result directly,

$$\frac{\ln^2(x)}{x}$$

with

$$u = \ln(x) \quad du = \frac{1}{x} dx$$

we get

$$\int \frac{\ln^2(x)}{x} dx = \int u^2 du = \frac{1}{3} u^3 + C = \frac{1}{3} \ln^3(x) + C.$$

Since

$$\lim_{x \rightarrow \infty} \frac{1}{3} \ln^3(x) = \infty$$

we have

$$\int_1^{\infty} \frac{\ln^2(x)}{x} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{\ln^2(x)}{x} dx = \infty$$

Example #9

Find

$$\int_1^{\infty} \frac{\sin(x) + 2}{x^2} dx.$$

Observe

$$0 < \sin(x) + 2 \leq 1 + 2 = 3$$
$$\Rightarrow 0 < \frac{\sin(x) + 2}{x^2} \leq \frac{3}{x^2}$$

Since

$$\int_1^{\infty} x^{-2} dx = 1 \Rightarrow \int_1^{\infty} \frac{3}{x^2} dx = 3$$

we conclude

$$\Rightarrow \int_1^{\infty} \frac{\sin(x) + 2}{x^2} dx \text{ converges}$$

by the Comparison Test.

Remark

One can show:

1. $\int_1^\infty 1/x^p \, dx$ converges if and only if $p > 1$.
2. $\int_0^1 1/x^p \, dx$ converges if and only if $p < 1$.
3. $\int_0^\infty 1/b^x \, dx$ converges for $b > 1$.
4. $\int_{-\infty}^0 b^x \, dx$ converges for $b > 1$.

You need to prove these on the exams.